HAXLR8 3.0

NATIONAL LEVEL HACKATHON • MIT MYSORE

SMART CITY
PROBLEM STATEMENTS

15 Challenges | Pick one. Build it. Power the city.

Smart city innovation beautifully intertwines technology and civic life, turning everyday urban failures in water, power, traffic, safety, and waste into opportunities to build solutions that make cities resilient, efficient, and livable.

1

Real-Time Urban Flooding Detection and Automated Drainage Intervention in Flood-Prone City Zones

Indian cities flood within minutes of monsoon bursts while drainage authorities learn about it only through citizen complaints, hours too late. Design a solution that detects waterlogging as it forms and triggers timely drainage response before traffic, property, and lives are affected.

2

Intelligent Streetlight Energy Optimization with Continuous Fault Detection Across Municipal Grids

Large shares of municipal streetlights stay dead for weeks because faults are discovered by accident, and functioning lights burn all night on empty roads. Design an intelligent solution that continuously detects failures and adapts streetlight operation to actual need, cutting both energy waste and dark streets.

3

Continuous Water Pipeline Leak Detection and Proactive Pressure Regulation in Aging Distribution Networks

Indian cities lose nearly a third of treated drinking water to undetected leaks in aging pipes, while many wards simultaneously face low pressure supply. Design a continuous monitoring solution that pinpoints leaks early and regulates distribution pressure to save water and stabilize supply.

4

Automated Waste Fill-Level Monitoring and Intelligent Collection Dispatch for Municipal Waste Fleets

Municipal waste trucks follow fixed daily routes regardless of whether bins are full or empty, wasting fuel while overflowing bins rot in busy market areas. Design an automated solution that knows bin fill status across the city and dispatches collection intelligently where and when it is actually needed.

5

Real-Time Air Quality Monitoring and Public Exposure Alerting Near Schools, Markets, and Construction Zones

Dangerous air pollution spikes strike specific neighborhoods without warning while city-wide averages hide the risk children and street vendors actually face. Design a real-time solution that detects hyperlocal air quality deterioration and warns exposed populations and authorities before harm accumulates.

6

Intelligent Parking Space Detection and Guidance in Congested Commercial Districts

Drivers in busy Indian commercial districts circle for thirty minutes or more hunting parking, creating a large share of local traffic congestion and emissions. Design an intelligent solution that guides drivers directly to available parking and reduces unnecessary cruising in congestion hotspots.

7

Automated Manhole and Sewer Overflow Monitoring to Prevent Urban Flooding and Fatalities

Open manholes and sewer surges kill citizens and flood roads every monsoon, yet municipal corporations discover problems only after accidents are reported. Design an automated monitoring solution that detects dangerous manhole and sewer conditions in real time and alerts response teams before tragedy occurs.

8

Real-Time Pedestrian and Cyclist Safety Detection at Unsignalized and High-Risk Intersections

Pedestrians and cyclists face the highest road risk at busy unsignalized crossings where no system today detects their presence or near-miss patterns. Design a real-time solution that identifies vulnerable-road-user conflicts as they happen and enables corrective intervention.

9

Intelligent Flood Barrier Control and Proactive Underpass Closure for Low-Lying Roads During Extreme Rainfall

Vehicle owners drown in underpasses every monsoon because closures happen only after water visibly rises and vehicles are already trapped. Design an intelligent solution that predicts underpass flooding ahead of time and enforces closure before water becomes life-threatening.

10

Unified Municipal Complaint Triage and Automated Resolution Tracking Across City Departments

Citizen complaints about roads, waste, water, and streetlights scatter across departments with no accountability, and most residents never learn whether their complaint was even seen. Design a unified triage solution that routes every complaint to the right department automatically and tracks resolution to completion.

11

Data-Driven Public Transport Reliability and Crowding Prediction for Urban Commuter Planning

Urban commuters abandon public transport because buses arrive unpredictably and peak-hour crowding makes travel unbearable, pushing more private vehicles onto already choked roads. Design a data-driven solution that predicts service reliability and crowding so commuters and transport authorities can plan around reality.

12

End-to-End Emergency Response Coordination with Intelligent Traffic Signal Priority for Ambulances and Fire Services

Emergency vehicles lose critical minutes at every signal in Indian traffic, and control rooms have no unified picture of incident, vehicle, and route. Design an end-to-end coordination solution that clears the path intelligently for emergency vehicles from dispatch to arrival.

13

Predictive Urban Heat-Island Mapping and Cooling Resource Allocation During Extreme Heatwaves

Heatwaves now kill more Indians than most natural disasters, yet cities lack street-level heat maps showing which neighborhoods become lethal and when. Design a predictive solution that maps urban heat risk in advance and directs cooling resources and public warnings to the most exposed areas.

14

Continuous City-Wide Utility Consumption Anomaly Detection for Water and Energy Loss Identification

Unbilled consumption, theft, and silent infrastructure losses drain municipal revenues while officials discover them only through year-end audits. Design a continuous anomaly-detection solution that identifies abnormal consumption patterns across the city and pinpoints loss locations for immediate verification.

15

Intelligent Disaster Evacuation Route Planning and Real-Time Shelter Allocation During Urban Emergencies

During floods, fires, and building collapses, panicked evacuation routes clog within minutes and families reach shelters that are already full. Design an intelligent solution that plans evacuation dynamically and allocates shelters in real time as disasters unfold.

Select your problem statement and register for HAXLR8 3.0

